Practical 1 - Basic C# Program and Console Output

```
using System;
namespace StudentDetailsApp
{
    0 references
    class Program
    {
        0 references
        static void Main(string[] args)
        {
            Console.WriteLine("--- Student Details ---");
            Console.WriteLine("Name: John Doe");
            Console.WriteLine("Roll Number: 101");
            Console.WriteLine("Course: C# Programming");
            Console.ReadLine(); // Pauses the console
        }
    }
}
```

```
--- Student Details ---
Name: John Doe
Roll Number: 101
Course: C# Programming
```

Practical 2 - Data Types and Operators

```
using System;
0 references
class Program
{
    0 references
    static void Main()
    {
        Console.Write("Enter first number: ");
        double num1 = Convert.ToDouble(Console.ReadLine());
        Console.Write("Enter second number: ");
        double num2 = Convert.ToDouble(Console.ReadLine());
        Console.WriteLine("\n--- Results ---");
        Console.WriteLine("Addition: " + (num1 + num2));
        Console.WriteLine("Subtraction: " + (num1 - num2));
        Console.WriteLine("Multiplication: " + (num1 * num2));
        Console.WriteLine("Division: " + (num1 / num2));
    }
}
```

```
Enter first number: 5
Enter second number: 2

--- Results ---
Addition: 7
Subtraction: 3
Multiplication: 10
Division: 2.5
```

Practical 3 - Student Marks and Grade Calculation.

```
using System;
0 references
class Program
{
    0 references
    static void Main()
    {
        Console.Write("Enter Math marks: ");
        int math = Convert.ToInt32(Console.ReadLine());
        Console.Write("Enter Science marks: ");
        int sci = Convert.ToInt32(Console.ReadLine());
        Console.Write("Enter English marks: ");
        int eng = Convert.ToInt32(Console.ReadLine());

        int total = math + sci + eng;
        double pct = (double)total / 3;

        Console.WriteLine($"Total: {total} | Percentage: {pct:F2}%");

        if (pct >= 80)
            Console.WriteLine("Grade: A");
        else if (pct >= 60)
            Console.WriteLine("Grade: B");
        else if (pct >= 40)
            Console.WriteLine("Grade: C");
        else
            Console.WriteLine("Grade: Fail");
    }
}
```

```
Enter Math marks: 84
Enter Science marks: 68
Enter English marks: 75
Total: 227 | Percentage: 75.67%
Grade: B
```

Practical 4 - Loop-Based Application.

```
using System;
0 references
class Program
{
    0 references
    static void Main()
    {
        Console.Write("Enter a number for its multiplication table: ");
        int num = Convert.ToInt32(Console.ReadLine());

        Console.WriteLine($"\nMultiplication Table of {num}:");

        for (int i = 1; i <= 10; i++)
        {
            Console.WriteLine($"{num} x {i} = {num * i}");
        }
    }
}
```

```
Enter a number for its multiplication table: 11

Multiplication Table of 11:
11 x 1 = 11
11 x 2 = 22
11 x 3 = 33
11 x 4 = 44
11 x 5 = 55
11 x 6 = 66
11 x 7 = 77
11 x 8 = 88
11 x 9 = 99
11 x 10 = 110
```

Practical 5 - Employee Class using OOP Concepts.

```
using System;
5 references
class Employee
{
    2 references
    public int EmployeeId { get; set; }
    2 references
    public string Name { get; set; }
    2 references
    public double Salary { get; set; }
    // Constructor
    2 references
    public Employee(int id, string name, double salary)
    {
        EmployeeId = id;
        Name = name;
        Salary = salary;
    }
    2 references
    public void DisplayDetails()
    {
        Console.WriteLine($"ID: {EmployeeId}, Name: {Name}, Salary: {Salary}");
    }
}
0 references
class Program
{
    0 references
    static void Main()
    {
        Employee emp1 = new Employee(101, "Alice Smith", 55000);
        Employee emp2 = new Employee(102, "Bob Jones", 62000);

        emp1.DisplayDetails();
        emp2.DisplayDetails();
    }
}
```

```
ID: 101, Name: Alice Smith, Salary: 55000
ID: 102, Name: Bob Jones, Salary: 62000
```

Practical 6 - Inheritance and Polymorphism.

```
using System;
3 references
class Employee
{
    2 references
    public double BaseSalary { get; set; } = 40000;
    3 references
    public virtual void CalculateSalary()
    {
        Console.WriteLine($"Employee Salary: ${BaseSalary}");
    }
}
2 references
class Manager : Employee
{
    1 reference
    public double Bonus { get; set; } = 15000;
    3 references
    public override void CalculateSalary()
    {
        double total = BaseSalary + Bonus;
        Console.WriteLine($"Manager Salary (with bonus): ${total}");
    }
}
0 references
class Program
{
    0 references
    static void Main()
    {
        Employee e = new Employee();
        e.CalculateSalary();
        Manager m = new Manager();
        m.CalculateSalary();
    }
}
```

```
Employee Salary: $40000
Manager Salary (with bonus): $55000
```

Practical 7 - Student Record Management using List.

```
using System;
using System.Collections.Generic;
5 references
class Student
{
    3 references
    public int RollNo { get; set; }
    3 references
    public string Name { get; set; }
}
0 references
class Program
{
    0 references
    static void Main()
    {
        List<Student> students = new List<Student>();
        // Add records
        students.Add(new Student { RollNo = 1, Name = "Alice" });
        students.Add(new Student { RollNo = 2, Name = "Bob" });
        // Display records
        Console.WriteLine("--- Student List ---");
        foreach (Student s in students)
        {
            Console.WriteLine($"Roll: {s.RollNo}, Name: {s.Name}");
        }
    }
}
```

```
--- Student List ---
Roll: 1, Name: Alice
Roll: 2, Name: Bob
```

Practical 8 - Exception Handling Application.

```
using System;
0 references
class Program
{
    0 references
    static void Main()
    {
        try
        {
            Console.Write("Enter Numerator: ");
            int num1 = int.Parse(Console.ReadLine());
            Console.Write("Enter Denominator: ");
            int num2 = int.Parse(Console.ReadLine());
            int result = num1 / num2;
            Console.WriteLine($"Result: {result}");
        }
        catch (DivideByZeroException)
        {
            Console.WriteLine("Error: Cannot divide by zero.");
        }
        catch (FormatException)
        {
            Console.WriteLine("Error: Please enter valid integer numbers.");
        }
        finally
        {
            Console.WriteLine("Execution of try-catch completed.");
        }
    }
}
```

```
Enter Numerator: 10
Enter Denominator: 2
Result: 5
```

Practical 9 - Windows Form – Simple Registration Form.

```
using System;
using System.Drawing;
using System.Windows.Forms;
namespace program9
{
    internal static class Program9
    {
        [STAThread]
        static void Main()
        {
            Application.EnableVisualStyles();
            Application.SetCompatibleTextRenderingDefault(false);
            Application.Run(new Form1());
        }
    }
    public class Form1 : Form
    {
        TextBox txtName, txtEmail;
        RadioButton rbMale, rbFemale;
        CheckBox cbReading, cbSports;
        ComboBox cmbCourse;
        Button btnRegister;
        public Form1()
        {
            Text = "Simple Registration Form";
            Size = new Size(450, 480);
            StartPosition = FormStartPosition.CenterScreen;

            Label heading = new Label();
            heading.Text = "Student Registration Form";
            heading.Location = new Point(80, 20);
            heading.Size = new Size(300, 30);
            heading.Font = new Font("Arial", 14, FontStyle.Bold);
            Controls.Add(heading);

            Label lblName = new Label();
            lblName.Text = "Name:";
            lblName.Location = new Point(40, 80);
            lblName.AutoSize = true;
            Controls.Add(lblName);

            txtName = new TextBox();
            txtName.Location = new Point(150, 75);
            txtName.Size = new Size(200, 25);
            Controls.Add(txtName);

            Label lblEmail = new Label();
            lblEmail.Text = "Email:";
            lblEmail.Location = new Point(40, 120);
            lblEmail.AutoSize = true;
            Controls.Add(lblEmail);
```

```
            txtEmail = new TextBox();
            txtEmail.Location = new Point(150, 115);
            txtEmail.Size = new Size(200, 25);
            Controls.Add(txtEmail);

            Label lblGender = new Label();
            lblGender.Text = "Gender:";
            lblGender.Location = new Point(40, 160);
            lblGender.AutoSize = true;
            Controls.Add(lblGender);

            rbMale = new RadioButton();
            rbMale.Text = "Male";
            rbMale.Location = new Point(150, 155);
            rbMale.AutoSize = true;
            Controls.Add(rbMale);

            rbFemale = new RadioButton();
            rbFemale.Text = "Female";
            rbFemale.Location = new Point(230, 155);
            rbFemale.AutoSize = true;
            Controls.Add(rbFemale);

            Label lblHobbies = new Label();
            lblHobbies.Text = "Hobbies:";
            lblHobbies.Location = new Point(40, 200);
            lblHobbies.AutoSize = true;
            Controls.Add(lblHobbies);

            cbReading = new CheckBox();
            cbReading.Text = "Reading";
            cbReading.Location = new Point(150, 195);
            cbReading.AutoSize = true;
            Controls.Add(cbReading);

            cbSports = new CheckBox();
            cbSports.Text = "Sports";
            cbSports.Location = new Point(250, 195);
            cbSports.AutoSize = true;
            Controls.Add(cbSports);

            Label lblCourse = new Label();
            lblCourse.Text = "Course:";
            lblCourse.Location = new Point(40, 240);
            lblCourse.AutoSize = true;
            Controls.Add(lblCourse);
```

```
        cmbCourse = new ComboBox();
        cmbCourse.Location = new Point(150, 235);
        cmbCourse.Size = new Size(200, 25);
        cmbCourse.DropDownStyle = ComboBoxStyle.DropDownList;

        cmbCourse.Items.AddRange(new object[]
        {
            "BCA", "BBA", "BSc", "MCA"
        });

        Controls.Add(cmbCourse);

        btnRegister = new Button();
        btnRegister.Text = "Register";
        btnRegister.Location = new Point(150, 295);
        btnRegister.Size = new Size(100, 35);
        btnRegister.Click += BtnRegister_Click;
        Controls.Add(btnRegister);
    }

    private void BtnRegister_Click(object sender, EventArgs e)
    {
        string name = txtName.Text.Trim();
        string email = txtEmail.Text.Trim();

        if (name == "" || email == "" ||
            cmbCourse.SelectedIndex == -1)
        {
            MessageBox.Show(
                "Please enter Name, Email and select Course.",
                "Warning",
                MessageBoxButtons.OK,
                MessageBoxIcon.Warning
            );

            return;
        }

        string gender = "Not selected";

        if (rbMale.Checked)
            gender = "Male";
        else if (rbFemale.Checked)
            gender = "Female";

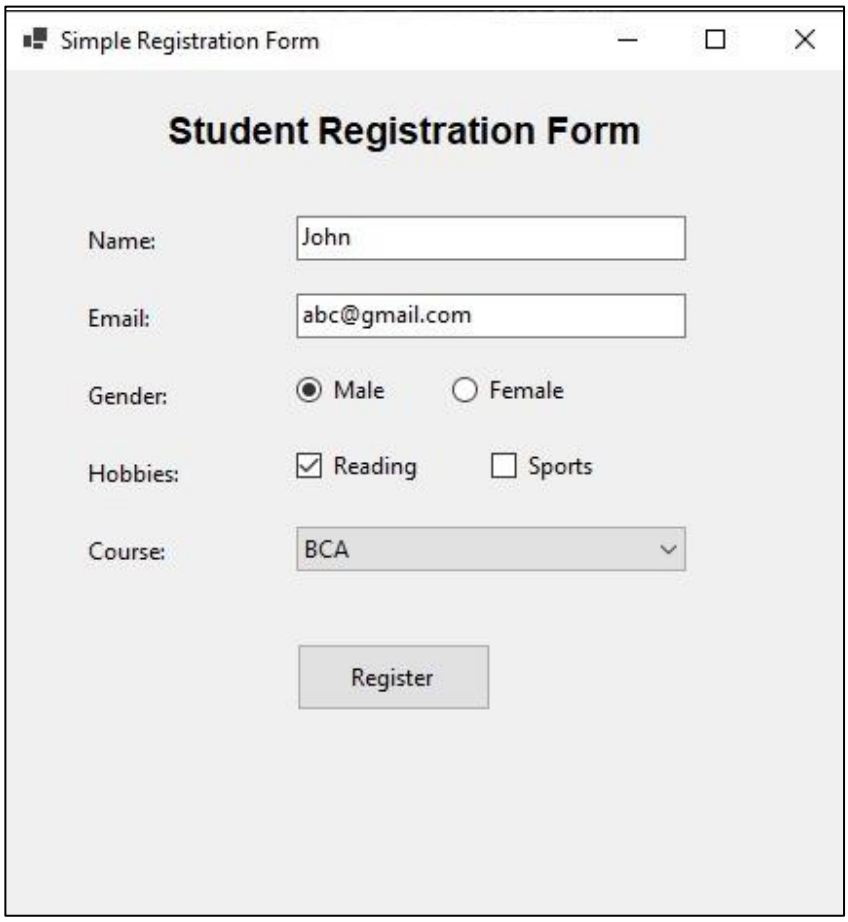
        string hobbies = "";
```

```
            if (cbReading.Checked)
                hobbies += "Reading ";

            if (cbSports.Checked)
                hobbies += "Sports ";

            if (hobbies == "")
                hobbies = "None";

            MessageBox.Show(
                "Registration Details\n\n" +
                "Name: " + name + "\n" +
                "Email: " + email + "\n" +
                "Gender: " + gender + "\n" +
                "Hobbies: " + hobbies + "\n" +
                "Course: " + cmbCourse.Text,
                "Registration Successful",
                MessageBoxButtons.OK,
                MessageBoxIcon.Information
            );
        }
    }
}
```


Simple Registration Form
Student Registration Form
Name:
John
Email:
abc@gmail.com
Gender:
Male
Female
Hobbies:
Reading
Sports
Course:
BCA
Register

Practical 10 - Database Connectivity using ADO.NET.

```
using System;
using System.Data;
using System.Data.SqlClient;
class Program
{
    static string cs = @"Data Source=(localdb)\MSSQLLocalDB;
                         Initial Catalog=UserDB;
                         Integrated Security=True;";

    static void Main()
    {
        CreateTable();

        while (true)
        {
            Console.WriteLine("\n===== USER DATABASE =====");
            Console.WriteLine("1. Insert User");
            Console.WriteLine("2. Display Users");
            Console.WriteLine("3. Update User");
            Console.WriteLine("4. Delete User");
            Console.WriteLine("5. Exit");
            Console.Write("Enter your choice: ");

            int ch = Convert.ToInt32(Console.ReadLine());
            switch (ch)
            {
                case 1:
                    InsertUser();
                    break;

                case 2:
                    DisplayUsers();
                    break;

                case 3:
                    UpdateUser();
                    break;

                case 4:
                    DeleteUser();
                    break;

                case 5:
                    return;

                default:
                    Console.WriteLine("Invalid choice!");
                    break;
            }
        }
    }
```

```
    static void CreateTable()
    {
        using (SqlConnection con = new SqlConnection(cs))
        {
            con.Open();

            string q = @"IF OBJECT_ID('Users', 'U') IS NULL
                         CREATE TABLE Users
                         (
                             Id INT IDENTITY(1,1) PRIMARY KEY,
                             Name VARCHAR(100)
                         )";

            SqlCommand cmd = new SqlCommand(q, con);
            cmd.ExecuteNonQuery();
        }
    }

    static void InsertUser()
    {
        Console.Write("Enter user name: ");
        string name = Console.ReadLine();

        using (SqlConnection con = new SqlConnection(cs))
        {
            con.Open();

            string q = "INSERT INTO Users (Name) VALUES (@name)";

            SqlCommand cmd = new SqlCommand(q, con);
            cmd.Parameters.Add("@name", SqlDbType.VarChar, 100).Value = name;

            int rows = cmd.ExecuteNonQuery();

            Console.WriteLine(rows + " row(s) inserted successfully.");
        }
    }

    static void DisplayUsers()
    {
        using (SqlConnection con = new SqlConnection(cs))
        {
            con.Open();

            string q = "SELECT * FROM Users";

            SqlCommand cmd = new SqlCommand(q, con);
            SqlDataReader dr = cmd.ExecuteReader();

            Console.WriteLine("\n---- User Records ----");
```

```
            while (dr.Read())
            {
                Console.WriteLine(
                    "ID: " + dr["Id"] +
                    " | Name: " + dr["Name"]);
            }

            dr.Close();
        }
    }

    static void UpdateUser()
    {
        Console.Write("Enter user ID to update: ");
        int id = Convert.ToInt32(Console.ReadLine());

        Console.Write("Enter new user name: ");
        string name = Console.ReadLine();

        using (SqlConnection con = new SqlConnection(cs))
        {
            con.Open();

            string q = "UPDATE Users SET Name=@name WHERE Id=@id";

            SqlCommand cmd = new SqlCommand(q, con);
            cmd.Parameters.Add("@name", SqlDbType.VarChar, 100).Value = name;
            cmd.Parameters.Add("@id", SqlDbType.Int).Value = id;

            int rows = cmd.ExecuteNonQuery();

            Console.WriteLine(rows > 0
                ? "User updated successfully."
                : "User ID not found.");
        }
    }
```

```
    static void DeleteUser()
    {
        Console.Write("Enter user ID to delete: ");
        int id = Convert.ToInt32(Console.ReadLine());

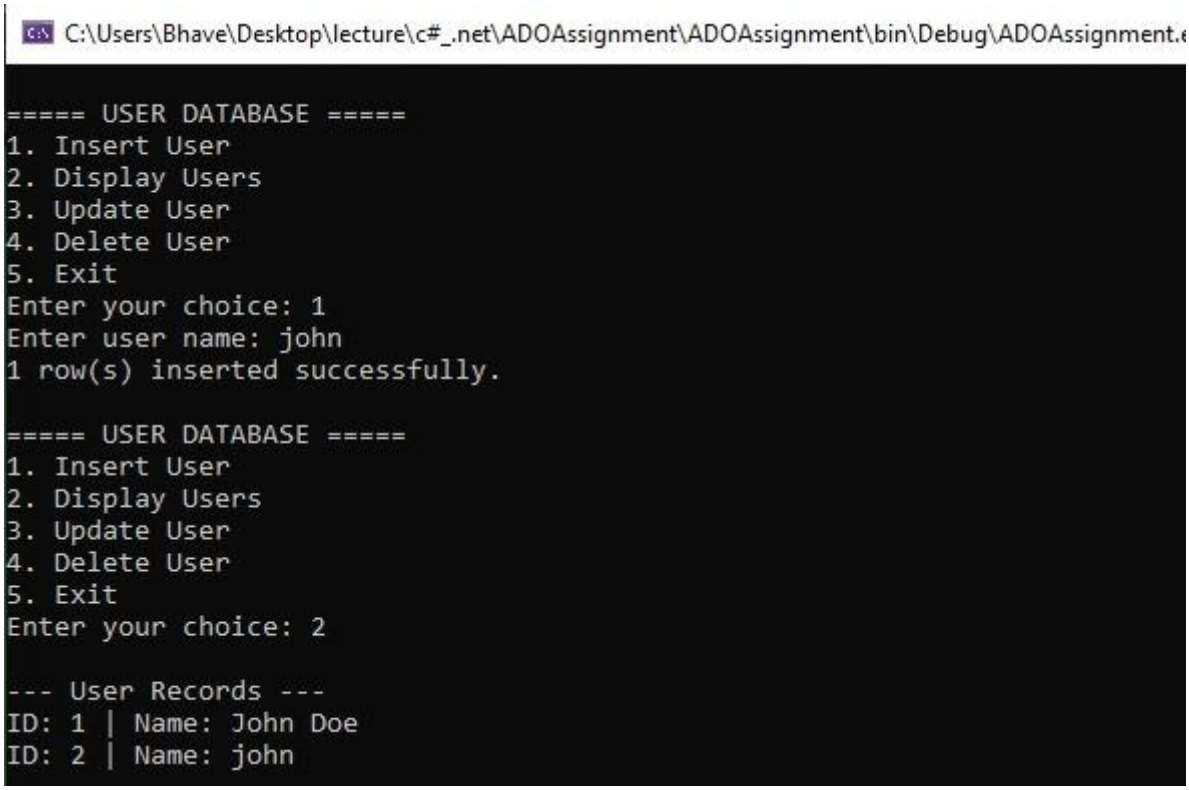
        using (SqlConnection con = new SqlConnection(cs))
        {
            con.Open();

            string q = "DELETE FROM Users WHERE Id=@id";

            SqlCommand cmd = new SqlCommand(q, con);
            cmd.Parameters.Add("@id", SqlDbType.Int).Value = id;

            int rows = cmd.ExecuteNonQuery();

            Console.WriteLine(rows > 0
                ? "User deleted successfully."
                : "User ID not found.");
        }
    }
}
```


C:\Users\Bhave\Desktop\lecture\c#_.net\ADOAssignment\ADOAssignment\bin\Debug\ADOAssignment.
===== USER DATABASE =====
1. Insert User
2. Display Users
3. Update User
4. Delete User
5. Exit
Enter your choice: 1
Enter user name: john
1 row(s) inserted successfully.
===== USER DATABASE =====
1. Insert User
2. Display Users
3. Update User
4. Delete User
5. Exit
Enter your choice: 2
--- User Records ---
ID: 1 | Name: John Doe
ID: 2 | Name: john